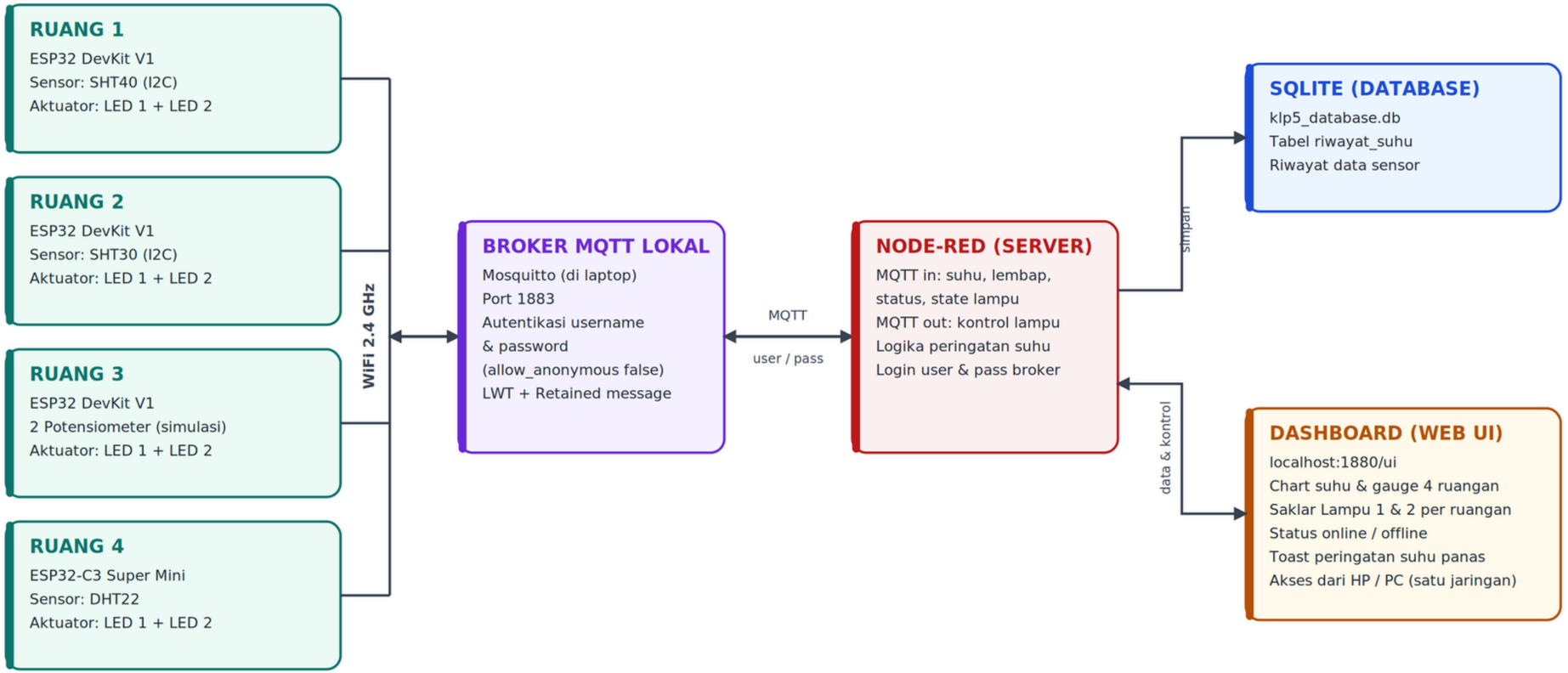
EVALUASI JARINGAN SENSOR & IOT

Sistem Monitoring dan Kontrol Multi-Ruangan Berbasis ESP32, MQTT, dan Node-RED

Kelompok 05

IKHTISAR SISTEM

Arsitektur Sistem Monitoring



Alur topic (prefix: pens/klp5/ruangN/, N = 1 sampai 4)

ESP32 → Broker → Node-RED

temperature, humidity (QoS 0, tiap 5 detik)
status: online / offline (LWT, retain)
lamp1/state, lamp2/state (konfirmasi lampu, retain)

Node-RED → Broker → ESP32

lamp1, lamp2 (payload "1" = ON, "0" = OFF, QoS 1)
Payload selain "0" / "1" diabaikan oleh ESP32

PROTOKOL KOMUNIKASI

Rancangan Topic Final

Topic Full (pens/klp5/ruangN/...)	Arah Data
.../temperature	ESP32 → Node-RED
.../humidity	ESP32 → Node-RED
.../lamp1	Node-RED → ESP32
.../lamp2	Node-RED → ESP32
.../status	ESP32 / LWT → Node-RED
.../lamp1/state	ESP32 → Node-RED
.../lamp2/state	ESP32 → Node-RED

FIRMWARE

Logika Kode ESP32

LWT

Publikasi status offline otomatis.

Reconnect

Pemulihan koneksi tanpa memblokir sistem.

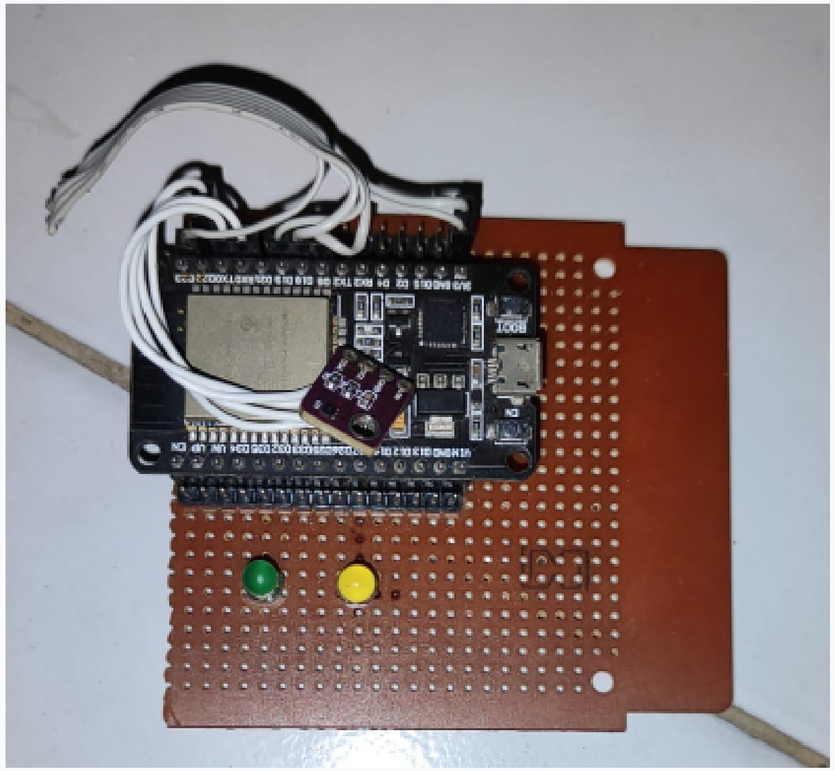
Callback

Eksekusi aktuator via payload MQTT.

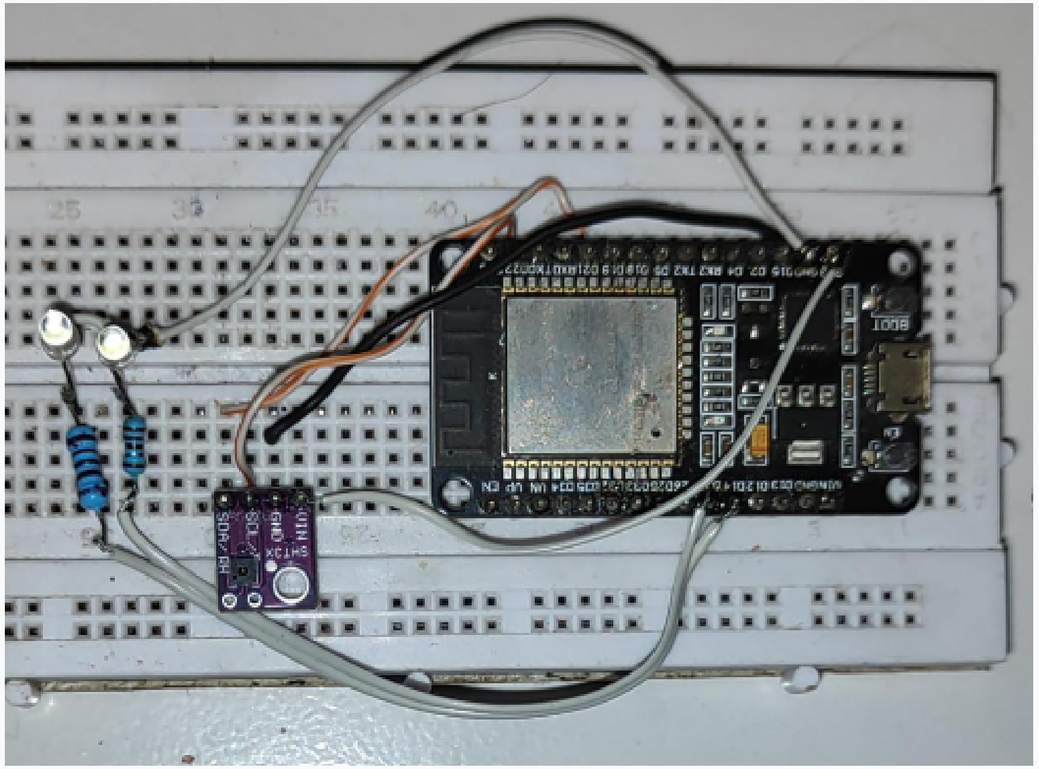
HARDWARE

Hardware Tiap Ruang

Ruang 1



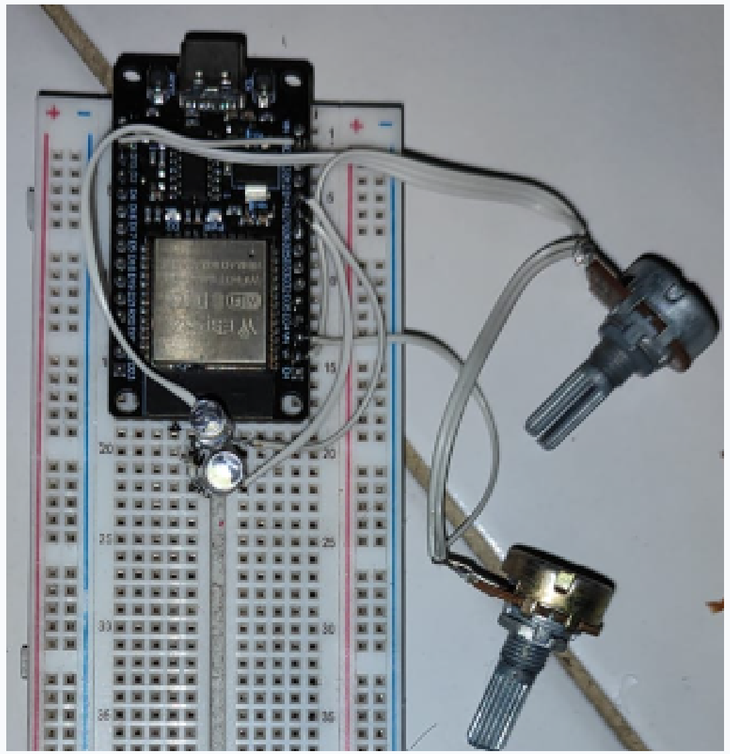
Ruang 2



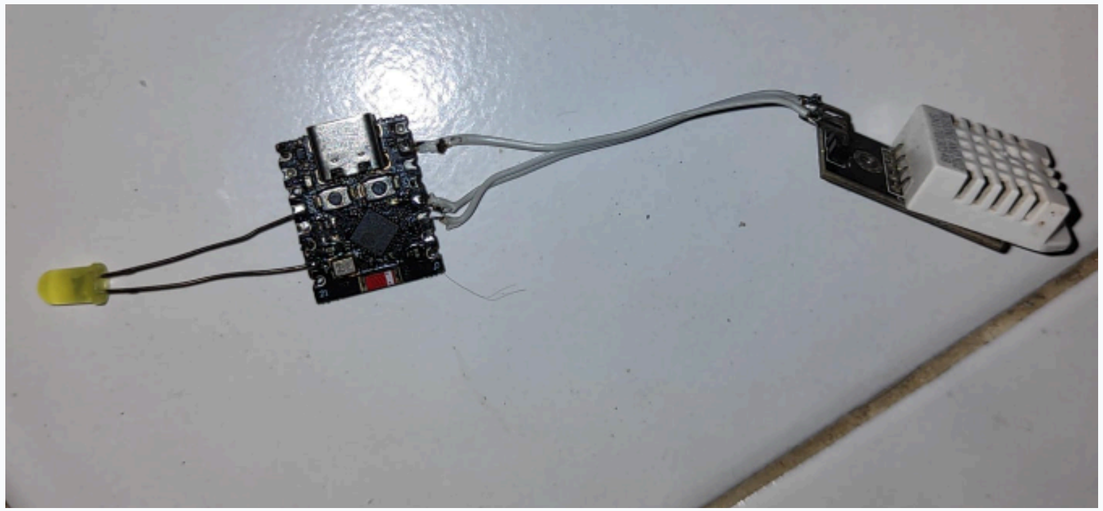
HARDWARE

Hardware Tiap Ruang

Ruang 3

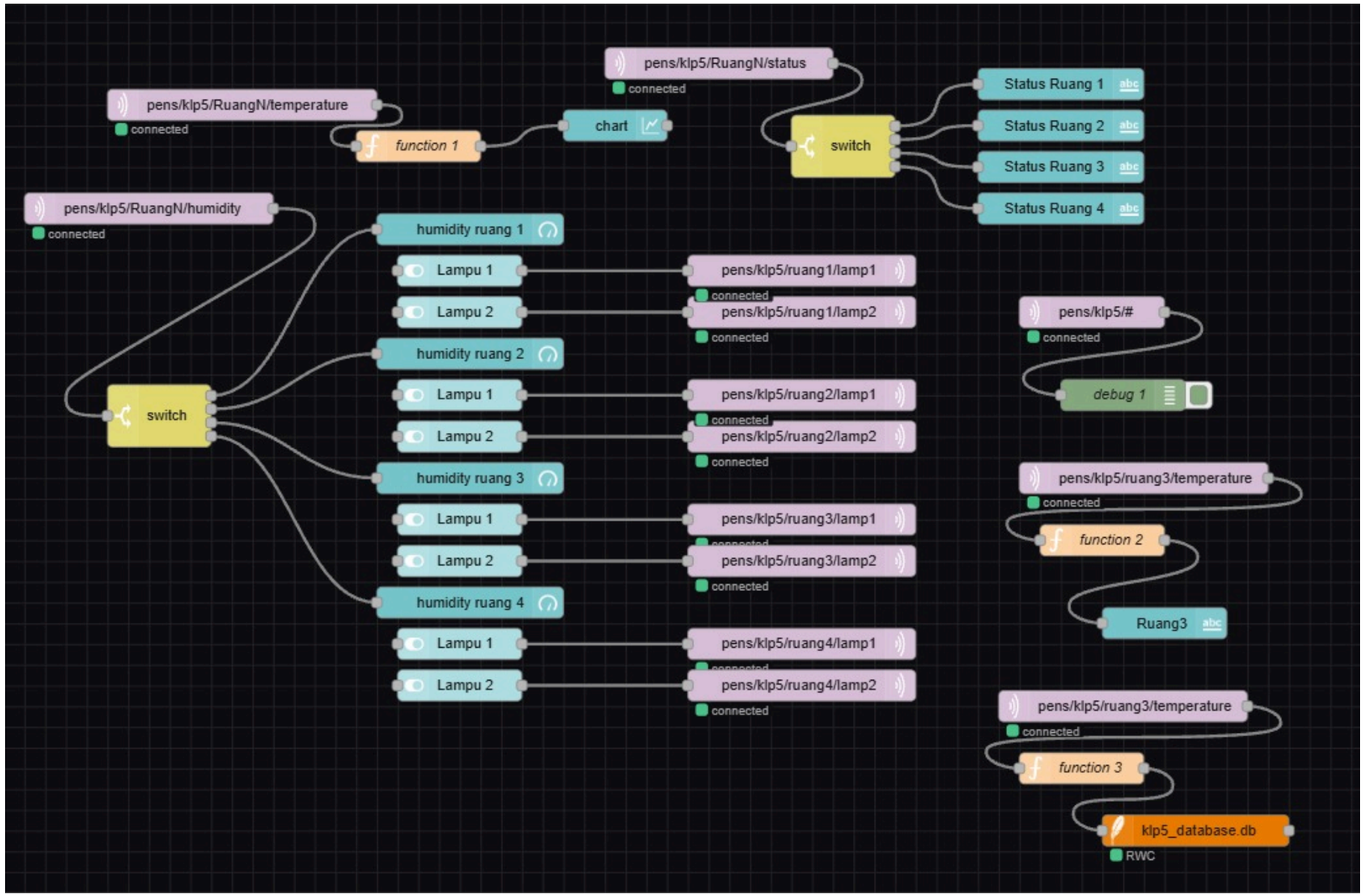


Ruang 4



BACKEND & DASHBOARD

Alur Node-RED



PEMECAHAN MASALAH

Kendala & Solusi

Kendala Status Lampu

UI tidak sinkron saat refresh.

Solusi: State Feedback

Implementasi Retain flag.

Kendala Logika Pin

Lampu built-in terbalik menyala.

Solusi: Pemisahan Modus

Fungsi khusus Active LOW/HIGH.

PEMECAHAN MASALAH

Kendala & Solusi

Node Red tidak terhubung ke broker

error Connection failed to broker

Solusi: Memasukkan Username dan Password

broker Mosquitto sudah diatur dengan allow_anonymous false

Teks pada menu navigasi samping UI

Teks UI sulit dibaca.

Solusi: Mengubah pengaturan Theme di panel Dashboard

mengubah Style menjadi Dark atau menggelapkan warna Font

NILAI TAMBAH

Nilai Tambah



Deteksi LWT

Identifikasi putus daya otomatis.



Toast Alerts

Peringatan ambang batas suhu.



State Feedback

Sinkronisasi data ke hardware.



Riwayat Data

Mentimpan data suhu dan kelembapan ke database



Terima Kasih

Tidak Membuka Sesi Tanya Jawab